
FORKSERVE: BRANCH-AWARE PREFILL PRUNING AND DISAGGREGATED KV FOR AGENTIC LLM SERVING

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ABSTRACT

Agentic LLM serving forks a shared trunk into short residuals and then keeps one committed path. Prefix caches can reuse that trunk only after the tokens exist and a hash hits, so the first fan-out still copies the trunk once per child. Prefill/decode disaggregation separates time-to-first-token from decode latency, yet the connector transfers the pages the prefiller computed, including residuals that the harness later aborts. Branch cost should be incurred only for the residual that is committed, and only at the minimum-cost stage at which rejection is admissible.

ForkServe makes the branch a serving object. A copy-on-write context tree turns `fork` into an alias of the parent’s KV pages and allocates frames only for the residual. Aborting a sibling releases those frames alone, so live occupancy collapses to the committed spine. On that tree, ForkServe decides for each residual whether GPU prefill runs and whether any KV crosses the disaggregated connector. The decision is a cascade of increasing cost: a block-hash probe reuses published pages, a draft score skips a repeated loop before a kernel starts, other low-score residuals are prefilled and decoded for a short probe that continues to the budget only if the generated prefix clears the threshold, and a transfer gate inserts only the miss tail of a residual that is kept. Decode attends that spine alone. Longest-common-prefix commit binds the prompt presented by the harness. We prove that speculative tokens never enter the sampler, so committed outputs are identical to a baseline that prefills only committed tokens. A two-class scheduler admits speculative prefill only from the budget left after committed requests meet their latency margins, and cancels it when the commit conflicts.

On Qwen3-14B GSM8K, ForkServe matches prefix-cache accuracy at $0.72\times$ the median peak KV, cuts modeled fan-out prefill by 74.72% relative to that cache, and raises the concurrency a fixed pool can sustain by 35.53%. On Qwen3-8B MATH-500, skipping every low-score residual drops any-finished accuracy from 20/32 to 13/32 at fan-out 16; a 128-token probe that continues only when the generated prefix clears the threshold restores 20/32 and cuts end-to-end time from 104.1 s to 72.2 s. On Qwen3-8B GSM8K, $\tau_{\text{pre}}=0.15$ in front of ESC, Speculative Rejection, or DPTS keeps 16/16 any-finished at $k=16$, cuts computed prefill by 23%, and reduces ESC time from 17.05 s to 14.14 s. $\tau_{\text{pre}}=0.45$ is faster at that width (10.93 s, still 16/16) but drops two questions at $k=4$ and $k=8$.

1 INTRODUCTION

LLM serving began as a single-turn problem: a prompt arrives, the engine prefills, and tokens are streamed [47, 18, 53]. Production agents—coding assistants [44, 39], computer-use systems [55], and multi-agent workflows [42, 15]—execute programs of LLM calls interleaved with tools [46, 50]. Tree-of-Thoughts and MCTS expand sibling thoughts that share a prefix and diverge in a short residual [45, 26]. The serving unit is a *branching execution*, not a request.

Prior systems raise the atom from a request to a program [25,

17], a session [20, 43], or a workflow DAG [36, 23, 48]. They share an implicit assumption: the next LLM call is a unique continuation of the current context. After a tool call is emitted, the next prompt is a *distribution* over branches the engine can see *before* the observation returns: known suffixes with $p=1$ (templates, wrappers), fan-out siblings that share a trunk of length L and differ in residuals ℓ_i , and observation residuals with $p<1$. Current engines wait until the chosen branch is fully materialized. Automatic prefix caching (APC) [18, 53] discovers sharing only after tokens exist. Prefill–decode disaggregation [54, 31] isolates TTFT from inter-token latency, but the connector transfers every KV page the prefiller computed. Once KV retention has removed trunk recompute [20, 43], the remaining TTFT term is residual-suffix prefill.

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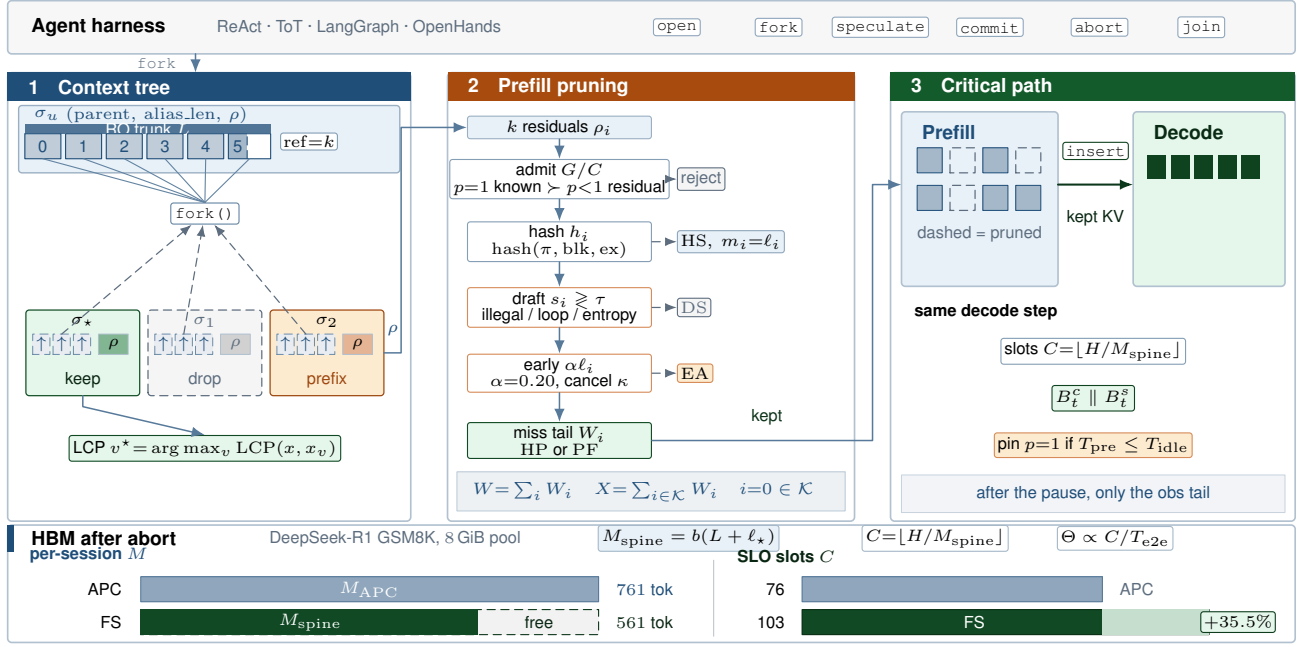


Figure 1. ForkServe architecture. (1) σ_u holds one read-only trunk; `fork` aliases it in $O(1)$; LCP binds the winner. (2) A residual exits at the first decisive action. $m_i = \ell_i$ is HS ($W_i=0$); $0 < m_i < \ell_i$ is HP; a clean low score is DS; the first $\alpha \ell_i$ failing is EA; otherwise PF prefills ℓ_i on the alias. Index 0 stays in $\mathcal{K}$, and X sums W_i only over that kept set. (3) Slots are $C = \lfloor H/M_{\text{spine}} \rfloor$; B_t^c runs beside B_t^s , and a known suffix is pinned when $p=1$ and $T_{\text{pre}} \leq T_{\text{idle}}$. After abort the live occupancy is the spine; the same pool admits 76 $\rightarrow$ 103 sessions.

This work. We present ForkServe, a serving system whose first-class object is the *branch* (Figure 1). Three mechanisms are coupled.

(1) A forkable context tree with copy-on-write KV pages (§3.2). Each session is a tree of committed and speculative nodes. `fork` creates a child that inherits the parent’s pages by reference; writes allocate residual blocks. `abort` drops a subtree and decrements residual refcounts only, so live occupancy collapses to the spine $M_{\text{spine}} = b(L + \ell_*)$.

(2) Cost-ordered prefill pruning and transfer gating (§3.4). On top of the tree, each residual ρ_i receives one action $a_i \in \mathcal{A}$ and a pair (W_i, X_i) (Figure 1). A published prefix of length m_i is retrieved: $m_i = \ell_i$ is HS ($W_i = 0$) and $0 < m_i < \ell_i$ is HP ($W_i = \ell_i - m_i$). A miss ($m_i = 0$) has no row in H ; DS and EA reject on $s_i < \tau$ before or inside the first $\alpha \ell_i$ tokens, and PF charges ℓ_i on the trunk aliased at `fork`. Index 0 is forced into $\mathcal{K}$, so the connector inserts only that kept miss tail [54, 31]. LCP commit binds the observed prompt; speculative tokens never enter the sampler.

(3) The spine on the critical path (§3.5). Decode slots are $C = \lfloor H/M_{\text{spine}} \rfloor$. On tick t the budget splits as $B_t = B_t^c + B_t^s$: B_t^c decodes x_* and B_t^s prefills ρ_*^+ in the same step. A known suffix with $p(w)=1$ and $T_{\text{pre}}(|w|) \leq T_{\text{idle}}(u)$ is

pinned during the tool pause; the observation arrives as the unmatched tail.

§4 reports four measurements that are not interchangeable. On Qwen3-14B GSM8K, ForkServe matches APC accuracy at $0.72\times$ median peak KV; fan-out is still slower than APC. A control-plane study of APP then shows that hash skip, draft prune, and the P/D gate remove most of that prefill on residuals that fail the draft score, and that the measured KV gap raises SLO-feasible concurrency by 35.5%. The GPU rule is narrower than that blanket skip: only a repeated loop is dropped before prefill, and any other low-score residual is probed and extended when the generated prefix clears the threshold (Table 6). On Qwen3-8B GSM8K, $\tau_{\text{pre}}=0.15$ keeps 16/16 any-finished at $k=8$ and $k=16$ while cutting computed prefill by 23% and ESC time from 17.05 s to 14.14 s; $\tau_{\text{pre}}=0.45$ is faster at $k=16$ but drops two questions at smaller fan-out (Tables 7 and 8). A Qwen3-8B GPU run of ForkServe+ shows the same KV ordering. A six-model forest ($n=40$) keeps GSM8K peak KV at $0.72\text{--}0.75\times$ APC.

Table 1. Notation for one branching step and for APP.

Symbol	Domain	Meaning
$L, \ell_i, \ell_\star$	$\mathbb{N}$	trunk; residual i ; committed residual
ρ_i, x_u	tokens	residual of child i ; committed trunk
h_i	digest	$\text{hash}(\text{parent}, \rho_i[0:P], \text{extra})$
m_i	tokens	full-page prefix of ρ_i in H , steps of P
s_i, s'_i, τ	$[0, 1]$	score of ρ_i ; score of the probe y_i ; threshold
t_0, D, d_i	tokens	probe length; decode budget; length assigned to i
α	$(0, 1]$	early-abort prefix fraction (0.20)
p_i, q_i	$[0, 1]$	admit probability; residual-prefix hit rate
$a_i \in \mathcal{A}$	enum	skip / partial / abort / prefill (Eq. 5)
$\mathcal{K}$	$\subseteq [k]$	residuals that remain after APP
W_i, X_i	tokens	GPU miss tail; connector payload
b, P	bytes, tok	KV bytes per token; page size
$B_t = B_t^c + B_t^s$	tokens	two-class batch budget on tick t

2 BACKGROUND AND MODEL

2.1 Prefill, Decode, and Idle

Autoregressive inference has two mechanically different phases [47, 54, 31]. *Prefill* consumes the prompt in parallel and is compute-bound; *decode* emits one token (or a small chunk) per step and is memory-bandwidth bound. PagedAttention [18] stores KV in non-contiguous pages; RadixAttention [53] indexes pages by token prefix so concurrent requests that share a system prompt hit a common trunk. Two consequences matter for agents. First, next-turn TTFT is the prefill of tokens that miss the cache, plus queuing. Second, a memory-bound decode batch rarely saturates the tensor cores that a concurrent chunked prefill could use [2]. Agent serving adds a third fact: the prompt of turn $i+1$ is a *delta* on turn i , applied after a tool pause of highly variable length [20, 43].

2.2 A Branching Step

We treat one agentic step as a rooted prefix tree, not as k independent completions.

Definition 1 (Branching step). A parent node u holds a committed trunk x_u of length L . A harness-declared fan-out produces k children with residuals ρ_i of length ℓ_i , $i \in \{1, \dots, k\}$. After a winner $\star$ is bound, only the spine $x_\star = x_u \parallel \rho_\star$ of length $L + \ell_\star$ remains live.

Table 1 is the symbol set used in the rest of the paper. Let b be bytes of KV per token (all layers, one sequence) and P the page size in tokens. Three occupancies follow immediately:

$$\begin{aligned} M_{\text{clone}} &= b(kL + \sum_i \ell_i), \\ M_{\text{CoW}} &= b(L + \sum_i \ell_i), \\ M_{\text{spine}} &= b(L + \ell_\star). \end{aligned} \quad (1)$$

The first-fan-out work of a scheme is the number of tokens sent to a GPU prefill kernel before the winner is known; the transfer span is the number of tokens a P/D connector ships.

Why APC misses the first fan-out. APC indexes a *full* block by $h = \text{hash}(\text{parent}, \text{tokens}, \text{extra})$ *after* those tokens exist [18]. A same-batch ToT expansion looks up k children before any child has published a hash, so the walk misses and the engine clones k trunks: $W_{\text{APC}} = kL + \sum_i \ell_i$. A later replay of the committed winner is a full hash hit ($W=0$). Hash-only prefill therefore helps the second session, not the first expand.

Why prefill–decode disaggregation does not raise tokens/s. DistServe and Splitwise isolate TTFT from inter-token latency by placing prefill and decode on different instances and moving KV through a connector (`insert` is non-blocking, `drop_select` blocks) [54, 31]. That baseline transfers every page the prefiler computed,

$$X_{\text{PD}} = kL + \sum_i \ell_i, \quad (2)$$

so first-fan-out transfer equals first-fan-out work. P/D is a placement, not a branch filter.

Four operators that generate the tree. Tool intercept [46, 34] idles the engine while the environment runs; a known suffix is available immediately and an observation residual arrives later. Fan-out/join [42, 15] launches k children that share x_u and differ in short role residuals. Search [45, 26] expands b siblings, scores them, and recurses on a shared spine. Context transformation [30] forks a summarized child while the original spine stays abortable. In all four cases the next call is drawn from a discrete set $\mathcal{B}(u)$; existing engines observe only the child that is eventually issued.

Insight 1 (Prune before prefill). APC reuses a trunk only after a hash hit; prefill–decode disaggregation transfers the span that was computed. The missing operation is a per-residual decision (W_i, X_i, a_i), issued *before* a GPU kernel and *before* `insert`.

Insight 2 (Fork by reference). On fan-out the trunk is typically several times the per-child residual. Materializing k independent KV copies allocates $\Theta(k)$ redundant HBM for tensors that attention reads as identical. A location-addressed alias at `fork` makes a miss residual-only.

3 DESIGN

ForkServe is placed between the agent runtime and the LLM engine (Figure 1). The harness announces structure; the control plane admits, prunes, binds, and retains; the backend moves pages and tokens.

3.1 API

<code>open(session)</code>	Create a root; prefill system+user.
<code>fork(u, bid, tok?)</code>	Child of u ; optional known suffix.
<code>speculate(v, pri)</code>	Mark v eligible for speculative prefill.
<code>commit(u, tokens)</code>	Bind by LCP; start decode.
<code>join(children, pol)</code>	Merge fan-out (all/first/kofn).
<code>abort(v) / close</code>	Drop a subtree; end the session.

`commit` is the only verb on the critical path of user-visible tokens.

3.2 Forkable Branch State

Definition 2 (Context tree). A session is a rooted tree $T = (V, E)$. Node u stores a token sequence x_u such that $x_{\pi(u)} \preceq x_u$, a residual $\delta_u = x_u[|x_{\pi(u)}|:]$, a mode $\rho_u \in \{\text{SPEC}, \text{COMMIT}, \text{IDLE}, \text{DEAD}\}$, a generation g_u , and a logical page table $\sigma_u = (\text{parent}, \text{alias_len}, \rho_u^{\text{pages}})$.

Invariants: (i) prefix; (ii) at most one committed child per node except during join; (iii) speculative nodes are invisible to sampling; (iv) a page’s refcount equals the number of non-dead nodes that map it. `fork(u, bid, xk)` copies two words ($\text{parent} \leftarrow \sigma_u$, $\text{alias_len} \leftarrow |x_u|$) and increfs the aliased frames— $O(1)$ in L . A write that hits $\text{ref} > 1$ or a read-only tail CoW-copies only the valid rows. Figure 2 is that pool. A multi-level fan-out is a *sequence* of CoWs (Figure 4): each edge aliases the parent and allocates only $+r$; live paths share every page on the common prefix; abort is residual-only, and a subtree with no live child cascades until the committed spine. The two compose: LCP commit publishes full winner pages into the APC index so a later session can hash-hit them (§3.4). Abort frees $b\ell_i$ bytes, not $b(L + \ell_i)$. TTL and offload are node-granular: speculative leaves first, then idle committed leaves, then the spine [20, 43].

3.3 Admission

Candidates $\mathcal{B}(u)$ come from harness-declared wraps, grammar mass over the next tool name (top- m), and a session-local prior of ($\text{parent_role}, \text{bid}$). Priors never invent a branch id. Each candidate is a pair $(x_b^k, \hat{x}_b^o)$: an exact known suffix and an optional observation guess. This is *input-side* speculation—a cache of a future prompt, not draft output tokens [19, 7].

The expected TTFT reduction and the interference-aware cost of a work item w are

$$G(w) = p(w) \cdot \min(T_{\text{pre}}(|w|), T_{\text{idle}}(u)) \cdot \eta(w), \quad (3)$$

$$C(w) = b \cdot |w| + \lambda \cdot \max(0, T_{\text{pre}}(|w|) - \gamma), \quad (4)$$

where η down-weights residuals by q_b and work that would finish after the pause, and λ matches 1 ms TBT to 4 ms TTFT [52].

Proposition 1 (Known-suffix first). *If two items have equal T_{pre} and w is a known suffix ($p=1, q=1$) while w' is an observation residual ($q < 1$), then $G(w)/C(w) \geq G(w')/C(w')$.*

Admission is a greedy fractional knapsack on G/C subject to idle horizon, residual HBM, branch cap m , and committed TBT margin. Chunks of size $\leq C_{\text{spec}}$ make a mid-prefill `commit` discard at most one in-flight chunk.

LCP commit. When the harness presents prompt x of parent u : (1) $v^* = \arg \max_v \text{LCP}(x, x_v)$ among live children; (2) pages covering the LCP become committed (CoW-split if mid-page); (3) the unmatched tail is a committed prefill; (4) sibling Spec nodes abort; (5) $g_u \leftarrow g_u + 1$ and decode is released. Speculative jobs tagged $g \neq g_u$ drop in $O(1)$.

Theorem 2 (Output identity). *For any session, the committed decode tokens of ForkServe equal those of a baseline that prefills each committed prompt from scratch (or from a prefix cache of committed tokens only), given equal seeds and weights.*

Proof sketch. Decode at a committed node u attends only to KV(x_u). That KV is produced by a committed prefill of x_u , or by a speculative prefix verified token-for-token by LCP plus a committed tail. Transformer KV of a prefix depends only on that prefix. Speculative nodes do not enter the sampler. $\square$

3.4 Cost-Ordered Prefill Pruning and Transfer Gating

Copy-on-write fixes trunk placement. It does not decide which residual enters a GPU kernel, nor which KV crosses the prefill–decode connector. APP makes both decisions per residual, (W_i, X_i) , by a cascade whose stage cost is strictly increasing (Figure 3). The stages are a block-hash probe, a CPU draft score, a prefix of the prefill kernel, the miss-tail prefill, and a connector `insert`. Stage $i+1$ runs only when stage i does not reject ρ_i . The committed winner is exempt: index 0 is forced into $\mathcal{K}$, speculative pages stay unpublished until LCP commit, and Theorem 2 is unchanged. CoW remains the storage substrate; APP is the filter on compute and on transfer. On the Qwen3-14B GSM8K forest, copy-on-write already stores less KV than APC, yet median fan-out is 140 ms against APC’s 87 ms: the k residuals are still submitted to prefill. The cascade is the stage that removes that work.

The cascade as an action. The order above is one action per residual. Write $\mathcal{A} = \{\text{HS}, \text{DS}, \text{EA}, \text{HP}, \text{PF}\}$ for hash skip, draft skip, early abort, hash partial, and prefill. Let m_i be the longest prefix of ρ_i covered by full pages of the published index H , so $m_i \in \{0, P, 2P, \dots\}$ and $m_i \leq \ell_i$. The probe key is $\text{trunk} \parallel \rho_i$, but the trunk itself is the alias

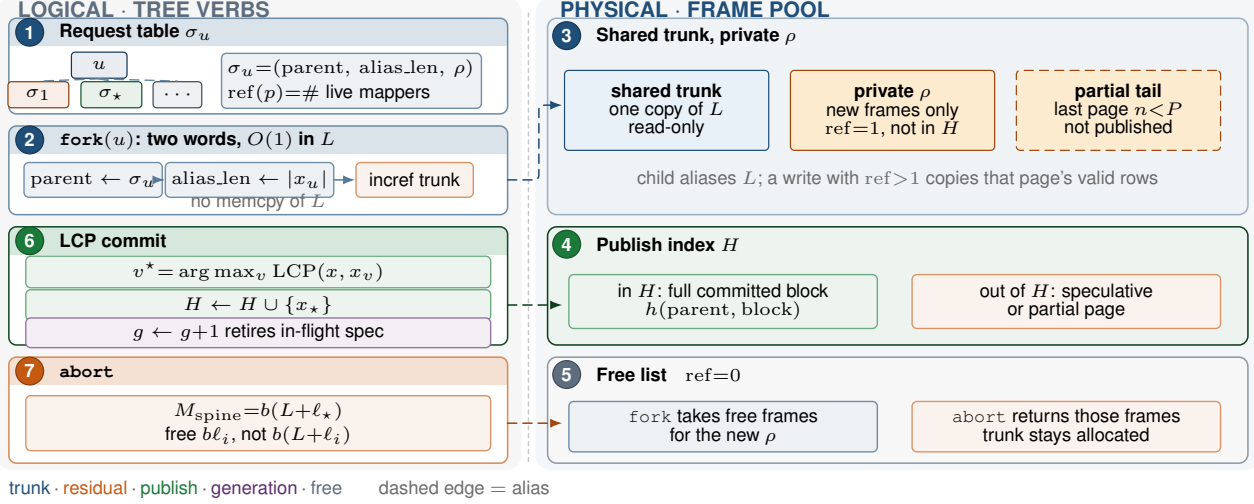


Figure 2. **Context-tree page pool.** `fork` aliases the trunk in $O(1)$ and allocates only ρ ①–③. LCP commit publishes x_* into H ④⑥. `abort` returns residual frames only, so live occupancy is the spine $M_{\text{spine}}=b(L+\ell_*)$ ⑤⑦.

written by `fork` ($\sigma_i \leftarrow (\sigma_u, |x_u|)$), so L is absent from W_i and $W_i \leq \ell_i$ on every action. Cases below are first-match.

$$a_i = \begin{cases} \text{HS} & m_i = \ell_i, \\ \text{HP} & 0 < m_i < \ell_i, \\ \text{DS} & m_i = 0 \text{ and } \rho_i \text{ is a repeated loop,} \\ \text{EA} & m_i = 0 \text{ and the first } \alpha \ell_i \text{ already fails,} \\ \text{PF} & \text{otherwise } (m_i = 0). \end{cases} \quad (5)$$

Winner $i=0$ is forced to $\mathcal{K}$. GPU work and connector payload are

$$W_i = \begin{cases} 0 & a_i \in \{\text{HS}, \text{DS}\}, \\ \lceil \alpha \ell_i \rceil & a_i = \text{EA}, \\ \ell_i - m_i & a_i = \text{HP}, \\ \ell_i & a_i = \text{PF}, \end{cases} \quad (6)$$

$$X_i = \begin{cases} W_i & i \in \mathcal{K} \wedge a_i \in \{\text{HP}, \text{PF}\}, \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

A full hit is a retrieval: $m_i = \ell_i$ reads published KV and issues no kernel ($W_i = X_i = 0$). A proper prefix retrieves m_i tokens and prefills the uncovered tail $\ell_i - m_i$. A miss ($m_i = 0$) has an empty probe. The draft score runs, and a residual that remains is a prefill of ℓ_i on the aliased trunk. Figure 5 is that rule on the residual “find the dragon.” The dragon head is the radix root on every row. A green prefix is a published page and is spliced onto decode; an orange tail is grown by prefill. Speculative pages stay out of H until LCP commit, so the first expand of an unpublished child has $m_i = 0$ and is the PF row.

Hash-local hits stay on the decode instance (the same published blocks are APC-visible there). Draft-skipped residuals never enter the buffer. The rejected length is a blanket skip. The corrected length is Algorithm 1.

$$d_i^\times = D \cdot [i = 0 \vee s_i \geq \tau]. \quad (8)$$

Every non-winner with $s_i < \tau$ had $d_i^\times = 0$: it was not prefilled and not decoded. That predicate drops residuals that still solve once generated. Table 2 remains the first-fan-out span under the actions of (5).

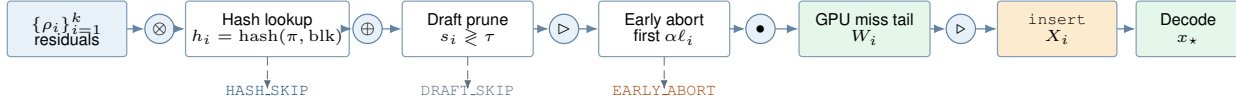
Lemma 3 (First-fan-out dominance). *On a cold index, $W_{\text{APP}} \leq W_{\text{CoW}} \leq W_{\text{APC}} = W_{\text{PD}}$ and $X_{\text{APP}} \leq X_{\text{PD}}$, with strict inequality in W whenever at least one repeated loop is draft-skipped, and strict inequality in X whenever $\mathcal{K} \neq [k]$ or any kept residual is a hash hit.*

Block-hash pruning. The first stage is a table probe and issues no model FLOPs. The index key is `hash(parent, block_tokens, extra)`, with `extra` carrying LoRA id, multimodal hash, or `cache_salt` on block 0 [18]. Only full pages enter H , which is why (5) tests m_i in steps of P . LCP commit inserts full blocks of x_* , which is what makes a later session a hash hit ($m_i = \ell_i$) rather than another fan-out. `fork` itself does not walk hashes: aliasing stays $O(1)$, and the lookup is a filter on the residual, not part of creating the child. A hash-local hit is also not a transfer. The same published blocks are already APC-visible on the decode instance, so $X_i=0$ even though the residual is kept.

Draft-score rejection. On a miss ($m_i = 0$ in (5)) the next stage is a CPU score, still before the prefill kernel. Each

ρ_i : i -th residual h_i : block hash s_i : draft score α : early-abort fraction κ : chunk W_i : GPU miss tail X_i : connector tokens a_i : action

(a) APP + P/D pipeline (Alg. 1)



Parallelism: k residuals Layout: location-addressed CoW, not content-addressed APC Winner $i=0$ is never dropped Spec pages unpublished
Work $W = \sum_i W_i$ Transfer $X = \sum_{i \in \mathcal{K}} W_i$ Occupancy after abort $M_{\text{spine}} = b(L + \ell_*)$ Replay of x_* : $W = X = 0$

(b) First-fan-out dataflow (same k, L, ℓ). Bottleneck in red.

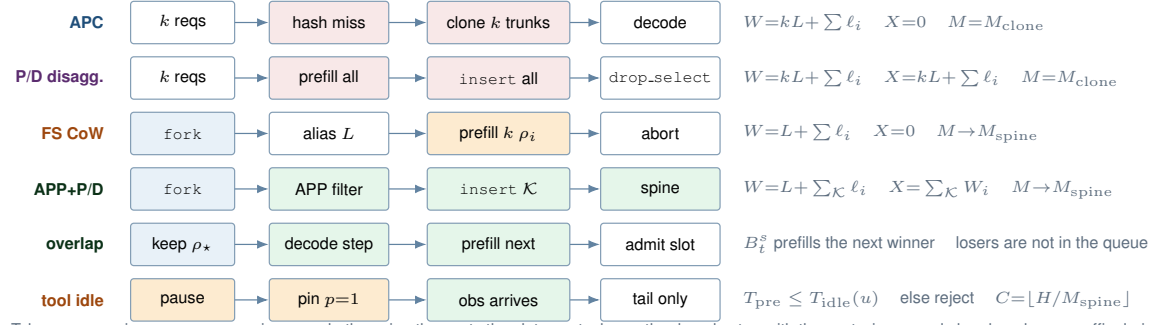


Figure 3. Cost-ordered prefill pruning and the critical path. (a) A residual exits at the first decisive stage. Only a kept miss tail reaches insert. (b) APC and baseline disaggregation clone the trunk. CoW aliases L . APP keeps the winner. The overlap row prefills that next winner in the decode step. The tool-idle row pins a known suffix during the pause and leaves the observation tail.

Table 2. Work, transfer, and memory spans (bottleneck in red). W_{pre} is GPU prefill tokens; X is connector tokens; M is live KV after abort; T_{ab} is abort cost in pages; replay is a later open of the published winner. PAR is the number of trunk copies touched on the first expand.

	W_{pre}	X	M after abort	T_{ab}	PAR _{trunk}	Replay W_{pre}
Recompute	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	$\Theta(kL/P)$	k	$kL + \ell_*$
APC (first expand)	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	LRU of hashed ρ	k	0
hash prefill (first)	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	LRU	k	0
P/D disagg.	$kL + \sum_i \ell_i$	$kL + \sum_i \ell_i$	$kL + \sum_i \ell_i$	full KV transfer	k	0
FS CoW	$L + \sum_i \ell_i$	0	$L + \ell_*$	$\Theta(\ell_i/P)$	1	L_{miss}
FS+ + P/D	$L + \sum_{i \in \mathcal{K}} \ell_i$	$\sum_{i \in \mathcal{K}} W_i$	$L + \ell_*$	lazy, residual only	1	0

$P=16$. Known-suffix items ($p=1$) are admitted before residuals ($p<1$). Spec pages are never hashed; LCP commit publishes full blocks of x_* .

residual receives a score from illegal-token and repetition checks, token entropy, or an optional draft model. An earlier draft of this rule treated every score below τ as a draft skip. That rule is not admissible. A residual the text score marks illegal still solves contest items once it is decoded, while a repeated loop solves none (Table 6). Draft skip is therefore only the repeated loop: that residual is not submitted to prefill and is not inserted as a hash key. The default prefill threshold is $\tau_{\text{pre}}=0.15$ for ESC, Speculative Rejection, and DPTS. Algorithm 1 is the correction of (??) at the decode bar τ . A residual in the loop set Λ is still (DS, 0). The winner, and every residual with $s_i \geq \tau$, is (PF, D). Every other non-winner is prefilled, decoded for t_0 tokens, and then assigned $d_i = D[s_i \geq \tau] + t_0[s_i < \tau]$. The probe is a prefix of decode, not the residual early-abort in panel (f) of Figure 5. On the MATH-500 measurements, $\tau=0.45$

and $t_0 \in \{64, 128\}$. The control-plane rows in §4.2 still charge the earlier blanket skip, which drops both low-score residuals before prefill.

Gated connector insert. Prefill–decode disaggregation places prefill and decode on separate instances [54, 31]. It does not choose a branch. The connector stays vLLM-shaped—insert is non-blocking, drop_select blocks—and a production pipe (Nixl, LMCACHE, Mooncake) keeps the same gate. Baseline disaggregation uses (2) and therefore transfers the computed span, including the k -way trunk. APP uses (7): X_i is nonzero only for a kept residual whose action is HP or PF, and then only for the miss tail that the kernel computed. Rejected residuals and hash-local hits stay on the prefill instance, so gated transfer reduces both W and X relative to baseline disaggregation. Let μ be



Figure 4. Staged CoW along many paths, threshold abort, cascade abort, shared prefixes. (a) Each edge is one CoW: the child aliases the parent’s pages and allocates only a residual (+r). Live paths A1b1x, A3a2x, and B2a1y share the trunk; the first two also share +r_A. Fork C has depth 2: at CoW-3 all four leaves die, so abort cascades through C1 and C2, then C; it stops at Root, which still has live children A and B. (b) KV stacks of the three live leaves, layers aligned: a shared layer is one physical page set with refcount > 1, not N copies. Aborting A1b1y or B2a1x frees only that leaf residual; cascade-aborting C also frees +r_{C1}, +r_{C2}, and +r_C.

Algorithm 1 Corrected admission, replacing (??). Λ is the repeated-loop set. y_i is a probe of t_0 tokens and s'_i is its draft score. Iverson brackets select the decode length. Only Λ keeps the old skip.

Input: $\{\rho_i\}_{i=0}^{k-1}$, threshold τ , probe t_0 , budget D
Output: action a_i , decode length d_i , set $\mathcal{K}$

```

1:  $\Lambda \leftarrow \{i \neq 0 : \text{Loop}(\rho_i)\}$ 
2: for  $i = 0$  to  $k - 1$  do
3:    $s_i \leftarrow \text{DraftScore}(\rho_i)$ 
4:   if  $i \in \Lambda$  then
5:      $(a_i, d_i) \leftarrow (\text{DS}, 0)$  {was  $d_i^\times = 0$  for all  $s_i < \tau$ }
6:   else if  $i = 0 \vee s_i \geq \tau$  then
7:      $(a_i, d_i) \leftarrow (\text{PF}, D)$ 
8:   else
9:      $y_i \leftarrow \text{Dec}(x_u || \rho_i, t_0)$ 
10:     $s'_i \leftarrow \text{DraftScore}(y_i)$ 
11:     $(a_i, d_i) \leftarrow (\text{PF}, D[s'_i \geq \tau] + t_0[s'_i < \tau])$ 
12:   end if
13: end for
14:  $\mathcal{K} \leftarrow \{i : a_i \neq \text{DS}\}$ 
15:  $W_i \leftarrow \ell_i \cdot [a_i = \text{PF}]$ 
16:  $\text{DecTokens} \leftarrow \sum_{i=0}^{k-1} d_i$ 

```

microseconds per transferred token. Fan-out time is

$$T_{\text{fan}} = T_{\text{pre}}(W) + \mu X + T_{\text{CoW}} + T_{\text{abort}}, \quad (9)$$

with T_{CoW} a pointer swap (no KV memcopy) and T_{abort} deferrable when abort is lazy. Under the control-plane model of §4.2 ($\ell \times 12 \mu\text{s}$, $\mu = 2 \mu\text{s}$), (8) is 139 ms for APC, 162 ms for baseline disaggregation, 47 ms for CoW, and 31.4 ms for APP.

3.5 The Spine on the Critical Path

After APP, fan-out is the prefill of the kept residual. End-to-end time is still $T_{\text{fan}} + T_{\text{dec}}$, and T_{dec} is the long term. Three placements put the spine on that sum. A fixed chunk width and a fan-out barrier leave the freed KV and the pruned queue off the path.

Slots follow the spine. Let H be the KV pool after weights. A live tree occupies $M_{\text{spine}} = b(L + \ell_*)$. APC still holds the aborted residuals until they expire, so its live set is the larger M_{APC} . The number of concurrent decode slots is

$$C = \lfloor H/M \rfloor, \quad (10)$$

with $M = M_{\text{spine}}$ for ForkServe and $M = M_{\text{APC}}$ for the prefix cache. C is not a constant chunk size. A request that finishes its current decode step releases the slot, and the next queued spine takes it. Throughput is

$$\Theta \propto C/T_{\text{e2e}}. \quad (11)$$

The same H therefore admits more sessions once occupancy is the spine, and the wall clock of a batch is the number of waves n/C , not n divided by a fixed width.

Prefill the next winner during decode. Each tick t has a token budget B_t from the chunked-prefill cap [2]. JIT-Serve’s margin [52] gives each committed request r a minimum token rate $\underline{s}_r$.

$$B_t^c = \min\left(B_t, \sum_{r \in R_t^c} \max(\underline{s}_r \Delta, 1)\right), \quad (12)$$

$$B_t^s = B_t - B_t^c. \quad (13)$$



Figure 5. Seven cuts of “find the dragon.” The dragon head is the radix root; the body is one residual. Green tokens are full pages in H and are spliced onto decode. Orange tokens are absent from H and are grown by prefill. Gray tokens are withheld. Chips stand in for tokens: a real hit ends on a full page, and each row is a different value of m_i . (a) The body is a hit (HS, $m_i = \ell_i$, $W_i = 0$). (b) The front half “dra” is in the tree and “gon” grows (HP, $W_i = \ell_i - m_i$). (c) The cut falls after “d”. (d) The cut falls before the last character. (e) A miss grows the body from the neck (PF, $m_i = 0$, $W_i = \ell_i$); the head stays the alias. (f) The first $\lceil \alpha \ell_i \rceil$ tokens grow and the kernel stops (EA). (g) A repeated loop is withheld (DS, $W_i = 0$). A low score that is not a loop is not this row: it is prefilled, then decoded for a probe (Algorithm 1).

B_t^c is committed decode. B_t^s is not spent on the $k-1$ residuals APP already rejected. It prefills the miss tail of the next winner, in the same step as the running decode. If $B_t^s = 0$, no speculative chunk is admitted. The fan-out of losers is not a generate that must finish before any decode token. A tree-sticky router pins the trunk to the worker that prefills the root; only residual work may migrate.

Hide the known suffix in the tool pause. A candidate with $p=1$ is an exact wrapper, not a guess. If $T_{\text{pre}}(|w|) \leq T_{\text{idle}}(u)$, that prefill runs while the tool is in flight, including when the parent is still marked committed. The pause is the deadline. A suffix that would finish after $T_{\text{idle}}(u)$ is rejected, as is any observation residual the pause cannot cover. When the observation arrives, LCP commit checks the pinned prefix and the committed prefill is the unmatched tail. Under P/D the same suffix is prefiller work tagged $\text{soft-deadline} = T_{\text{idle}}(u)$ and is inserted once, onto the trunk owner. Speculative tokens still do not enter the

sampler, so Theorem 2 is unchanged.

4 EVALUATION

The GPU measurements share one node and one engine, and they differ by model, tensor-parallel width, and workload. §4.1 is the full GSM8K forest on Qwen3-14B at $\text{tp}=2$. §4.2 is a control-plane cost model of APP, with no GPU, against APC, hash prefill, baseline prefill–decode disaggregation, ForkServe, and three decoding-time pruners. §4.2 measures the GSM8K prefill threshold $\tau_{\text{pre}}=0.15$ on Qwen3-8B and the accuracy–latency trade-off when that rule sits in front of ESC, Speculative Rejection, and DPTS. §4.3 runs ForkServe+ on Qwen3-8B at $\text{tp}=2$ and $\text{tp}=4$. §4.4 is a three-turn Tree-of-Thoughts on Qwen3-4B and Qwen3-8B, one problem per generate. §4.5 repeats the forest on six families at $n=40$. §4.6 sweeps branching, depth, pruning, and batch width on Llama-3-8B, Mistral-7B, and DeepSeek-R1-Distill-Llama-8B, then runs a 2048-token contest forest

Table 3. GSM8K test, Qwen3-14B, $tp=2$, $n=1319$. Median over 165 batches of eight. Best storage in green.

	recompute	APC	FS
Accuracy	0.847	0.844	0.845
Peak KV, median (tok)	4136	1430	1030
Peak KV, max (tok)	4944	1632	1232
Peak / APC	$2.89\times$	$1.00\times$	$0.72\times$
Fan-out, median (ms)	326	87	140
Generation (min)	—	18.6	18.9
KV saving vs. rec.	—	0.65	0.75

on that DeepSeek model and on Qwen3-4B.

Setup. GPU runs use vLLM on four A100-SXM4 40GB GPUs, with `bfloat16` weights and $tp=2$ or 4. The models are Qwen3-14B, Qwen3-8B, Qwen3-4B, Qwen2.5-Math-1.5B, Llama-3-8B, Mistral-7B, and DeepSeek-R1-Distill-Llama-8B. Tree-of-Thoughts uses $k=4$ and a 512-token budget (2048 on the contest forest), in batches of eight. The three-turn run in §4.4 issues one problem per generate. A math answer is correct when its final number matches the reference.

4.1 GPU forest

Setup. This section uses Qwen3-14B at $tp=2$. The GSM8K test set has 1319 problems. Each problem is expanded with four thoughts and decoded for at most 512 tokens, in 165 batches of eight. Accuracy is the fraction of problems whose final number matches the reference.

Storage and expand latency. Table 3 reports the storage result and the expand-latency gap that APP is designed to close. Median peak KV is $0.72\times$ APC and $0.25\times$ recompute, and accuracy stays within one point of both. Storage and fan-out do not share that ordering: median fan-out is 140 ms for ForkServe and 87 ms for APC, because copy-on-write and abort execute on the expand path, whereas an APC hash hit does not. End-to-end generation remains within 2% of APC (18.9 vs. 18.6 minutes). The same peak ordering, $0.58\text{--}0.80\times$ APC, holds on seven math sets for Qwen3-14B, Qwen3-8B, and DeepSeek-R1-Distill-Llama-8B at $tp=2$ and $tp=4$. The peak is unchanged when the tensor width doubles; it follows the committed path, as (1) requires.

4.2 Prefill pruning

Setup. Control plane, no GPU. Ten sessions, $k=4$, $L=256$, $\ell=32$. Two of four residuals fail the draft score; the winner is index 0. Prefill time is $\ell \times 12 \mu\text{s}/\text{token}$; P/D transfer is $2 \mu\text{s}/\text{token}$. APC and hash prefill publish blocks only after a request materializes them, so the first fan-out clones k trunks. Baseline disaggregation uses that prefill and then transfers the live token span. ForkServe aliases the trunk by pointer swap and prefills the k residuals. APP

Table 4. Control-plane fan-out (10 sessions, $k=4$, $L=256$, $\ell=32$). Bottleneck in red; ForkServe+ in green. Not GPU wall-clock.

Phase	Method	W	T_{fan}	X	peak	skip	prune
fan-out	APC	11520	139.0	0	11520	0	0
	hash	11520	139.0	0	11520	0	0
	P/D disagg.	11520	162.1	11520	11520	0	0
	FS	3840	47.1	0	3840	0	0
	ForkServe+	2592	31.4	32	2880	9	30
	ForkServe+ / APC	−77.5%	−77.4%	—	−75%	—	—

Table 5. Decoding-time pruning on the Table 4 fan-out (10×4 , $L=256$). Decode-to-cut is the prefix that must be generated before a child can be dropped. “On losers” counts that prefix on the two low-score children. Same $12 \mu\text{s}/\text{token}$ clock. Not GPU wall-clock.

	prefill	decode	on losers	peak	T_{cut}
ESC	2560	20480	10240	23040	276.5
Spec. Rejection	2560	10240	5120	12800	153.6
DPTS	2560	4000	2000	6560	78.7
ForkServe+	2592	0	0	2880	31.4

adds hash skip, draft prune, early abort, lazy abort, and a connector that inserts only retained residuals.

Fan-out. Table 4 is the comparison the GPU gap motivates. On the first expand, hash prefill matches APC: the blocks are unpublished, so the hash probe does not remove the cloned trunks. Baseline disaggregation exceeds APC (162 vs. 139 ms) because it transfers that clone. ForkServe removes the clone by aliasing the trunk (3840 prefill tokens, 47 ms). APP then withholds prefill from 30 of the 40 residuals, which reduces T_{fan} to 31.4 ms and the connector payload from 11520 tokens to 32.

Decoding-time pruning. Table 5 runs ESC [21], Speculative Rejection [35], and DPTS [12] on this same fan-out. Each method prefills the shared trunk once. The length of the decode that produces the prune score is the length published for that method. ESC stops after a window of 5 agreeing samples; this fan-out has four children and two low-score answers, so the window stays open and every child is decoded to the 512-token budget. Speculative Rejection is charged its partial-reward horizon $\tau=256$ and then drops the lower half ($\alpha=0.5$); the memory-limit trigger in that paper would not fire at $k=4$, so τ is the earlier checkpoint. DPTS early-stops after its 100-token mini-step. The two low-score children are the ones removed, and the winner is kept. Decoded tokens use the same $12 \mu\text{s}/\text{token}$ clock. DPTS is the earliest cut of the three and still decodes 4000 tokens before the drop, 2000 of them on losers, reaching the cut at 78.7 ms. APP has already withheld those residuals: 0 tokens on losers and $T_{\text{fan}}=31.4$ ms. That row is the blanket skip. §4.2 replaces it: only a repeated loop is withheld before prefill.

Table 6. MATH-500 level ≥ 4 , Qwen3-8B, $n=32$, $D=1024$. Any-finished is exact match on a branch that ran the full budget. “Drop $s < \tau$ ” withholds every low-score residual before prefill. “Grow t_0 ” is Algorithm 1: loops stay at $d_i=0$; every other low-score residual is probed for t_0 tokens and continues to D only if $s'_i \geq \tau=0.45$. Time is trunk warm plus fan-out, in seconds.

	$k=4$	$k=8$	$k=16$
Full fan-out	12/32, 25.3	15/32, 52.5	20/32, 104.1
Drop $s < \tau$	6/32, 19.1	11/32, 25.6	13/32, 52.1
Skip loops only	11/32, 23.0	17/32, 41.5	16/32, 71.4
Grow $t_0=64$	8/32, 24.0	16/32, 42.0	18/32, 72.5
Grow $t_0=128$	9/32, 24.7	13/32, 43.3	20/32, 72.2

Probe-then-grow. The blanket skip is the wrong cut once the residual is not a loop. We measure the rule in Algorithm 1 on Qwen3-8B, one GPU each, four A100-40GB, $\text{tp}=1$. The slice is the first 32 MATH-500 problems at level 4 or 5. Temperature is 0, EOS is ignored, prefix caching is on, and the decode budget is $D=1024$. Half of the k residuals are contest strategies. The other half alternates a repeated loop with illegal text. A question counts when some branch that ran all 1024 tokens matches the gold answer. The winner is residual 0.

Table 6 is the correction. Dropping every residual below τ cuts any-finished from 12/32 to 6/32 at $k=4$ and from 20/32 to 13/32 at $k=16$. Loop branches are 0 for 0 at every k , so skipping them before prefill is safe. At $k=16$, a 128-token probe extends every illegal residual (128/128) and none of the loops. Any-finished returns to the full fan-out, 20/32, while end-to-end time falls from 104.1 s to 72.2 s and decode tokens from 524288 to 393216. Probing the loops as well extends none of them and costs 74.4 s for the same 20/32. At $k=8$ the shorter probe is the one that holds accuracy: grow at $P=64$ is 16/32 in 42.0 s, against 15/32 in 52.5 s for the full fan-out. At $k=4$ the probe does not recover the full fan-out (9/32 against 12/32); skipping the single loop, and decoding the illegal residual to D , is 11/32 in 23.0 s.

Accuracy–speed trade-off. The MATH-500 probe still uses a decode threshold $\tau=0.45$ on the generated prefix. Prefill admission is a different score, with default $\tau_{\text{pre}}=0.15$: it drops only the repeated loop (0.02) and starts illegal text (0.22) and every live strategy (1.0). We measure that rule on Qwen3-8B GSM8K, $n=16$, $D=512$, the same hopeless mix, four A100-40GB, $\text{tp}=1$. The host decoder is ESC, so a withheld child never pays residual prefill or a decode slot. A denser sweep of τ_{pre} collapses to three admission sets; 0.15 and 0.45 are the two nonzero operating points.

Table 7 is the trade-off. $\tau_{\text{pre}}=0.15$ keeps 15/16, 16/16, 16/16 and still cuts wall-clock time, more as the tree widens (−5%, −6%, −17%). Computed prefill tokens fall from 1320/2624/5008 to 1018/2020/3848. Raising τ_{pre} to 0.45 is the extra-speed point: −9%, −21%, −36% end-to-end,

Table 7. GSM8K, Qwen3-8B, $n=16$, $D=512$, ESC only. Any-finished is exact match on a full-budget branch. $\tau_{\text{pre}}=0.15$ drops the loop; 0.45 also drops illegal text. Time is trunk warm plus fan-out, in seconds. Computed prefill falls 23% at 0.15 and 45% at 0.45 at every k .

	$k=4$	$k=8$	$k=16$
ESC	16/16, 8.65	16/16, 10.84	16/16, 17.05
APP $\tau_{\text{pre}}=0.15$	15/16, 8.22	16/16, 10.19	16/16, 14.14
APP $\tau_{\text{pre}}=0.45$	14/16, 7.89	14/16, 8.60	16/16, 10.93

Table 8. Same GSM8K slice, $k=16$, $\tau_{\text{pre}}=0.15$ in front of three host decoders. Admitted children and computed prefill tokens are identical across hosts: prefill admission does not read τ , d_{DPTS} , or α . Any-finished stays 16/16.

	adm.	prefill	e2e (s)	any
ESC	16	5008	17.05	16/16
ESC+APP	12	3848	14.14	16/16
Spec. Rejection	16	5008	13.14	16/16
SR+APP	12	3848	12.47	16/16
DPTS	16	5008	11.86	16/16
DPTS+APP	12	3848	11.49	16/16

and 732/1448/2768 computed prefill tokens. It drops two questions at $k=4$ and $k=8$. At $k=16$ both thresholds stay at 16/16, so 0.45 is the faster matched-accuracy setting (10.93 s vs. 14.14 s). The default is 0.15 because it holds accuracy at every k .

Table 8 stacks the same $\tau_{\text{pre}}=0.15$ in front of each host. ESC, Speculative Rejection, and DPTS alone all report 5008 computed prefill tokens: a decode-time cut still prefills every child. Adding APP admits 12 of 16 residuals and 3848 prefill tokens on every host. Wall-clock falls 17% on ESC, 5% on Speculative Rejection, and 3% on DPTS, because those last two hosts would have stopped the loop later anyway. The accuracy that Table 7 preserves is unchanged: every row in Table 8 is 16/16. At $k=8$ the same threshold is 16/16 on ESC (10.84 to 10.19 s) and lifts Speculative Rejection and DPTS from 15/16 to 16/16.

Concurrency from measured KV. Our empirical results on DeepSeek-R1 GSM8K (761 APC tokens, 561 ForkServe tokens, $n=16$, $k=4$) in an 8 GiB pool in (10) give 76 concurrent trees for APC and 103 for ForkServe. Figure 6 sweeps offered QPS: under a 1 s P99 TTFT constraint the same model admits 384 QPS for APC and 512 for the pruned tree, and saturated decode throughput rises from 48640 to 65920 tokens/s (+35.5%). This uses measured M , not the control-plane peak of Table 4.

4.3 GPU ForkServe+

Setup. Qwen3-8B, bfloat16 , $n=80$, $D=512$, $\text{tp}=2$ and 4 on four A100-SXM4 40GB GPUs. ForkServe+ enables lazy abort, pointer swap, hash skip, draft prune, and the P/D gate. GSM8K and Game24 are ToT ($k=4$); Hu-

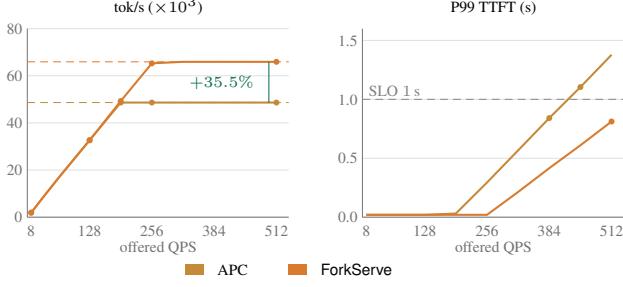


Figure 6. **SLO sweep** from measured DeepSeek GSM8K peaks. Left: saturated tok/s plateaus at $C \times \text{decode-rate}$ (76 vs. 103 slots). Right: APC crosses the 1 s P99 line at 448 QPS; ForkServe stays under it through 512.

Table 9. Qwen3-8B GPU forest, $n=80$, $D=512$. Peak KV is independent of tp . Game24 / HumanEval peaks stay with the spine.

	peak	D^+	e2e	acc.	time reduction	G24	HE
APC $tp=2$	1582	40960	36.0	0.900	—	1289	1970
ForkServe+ $tp=2$	1182	13356	28.2	0.863	0.78 $\times$	793	1882
APC $tp=4$	1582	40960	22.7	0.938	—	1289	1970
ForkServe+ $tp=4$	1182	12826	16.9	0.925	0.75 $\times$	793	1882

manEval is ReAct wrap+recovery ($k=2$).

Peak KV on this forest is 1182 against APC 1582 (0.75 $\times$) at both widths. Game24 and HumanEval peaks are 0.62 $\times$ and 0.96 $\times$ APC.

4.4 Multi-turn Tree-of-Thoughts

Setup. Qwen3-4B and Qwen3-8B, $bfloat16$, $tp=2$, temperature 0, $n=16$. Each problem is one session of $k=4$ agents and three turns, and each system issues that problem as its own generate. Turns 1 and 2 decode a 32-token step; the losers abort; the winner spine is the parent of the next fan-out. Turn 3 is the published answer, $D=512$ on GSM8K and $D=256$ on Game24.

Depth keeps one spine. Table 10 and Figure 8 are GSM8K and Game24 under that schedule. Peak KV is 275 against APC 318 on GSM8K (0.86 $\times$) and 216 against 265 on Game24 (0.82 $\times$), on both models. ForkServe stays within 6% of APC end-to-end (1.00–1.06 $\times$). Across 16 problems, three turns, and four agents, the expand issues 192 residuals. ForkServe+ prefills 48 of them and withholds the other 144 before they reach the GPU.

Table 11 and Figure 9 separate that expand from the winner decode. The expand speedup is 5.0–6.4 $\times$ for ForkServe and 7.9–10.6 $\times$ for ForkServe+. Decode does not match it. ForkServe decode is slower than APC because the two step decodes are timed there, which is why end-to-end stays at 1.00–1.06 $\times$. GSM8K accuracy matches APC on Qwen3-

Table 10. Three-turn ToT, $n=16$, $k=4$, $tp=2$, one problem per generate. End-to-end time is seconds per problem. Peak KV is tokens on the live spine. FS is ForkServe; FS+ is ForkServe+. Shortest end-to-end time and lowest peak in green.

	Qwen3-4B			Qwen3-8B		
	APC	FS	FS+	APC	FS	FS+
GSM8K acc.	0.938	0.938	0.938	1.000	0.938	0.938
GSM8K e2e (s)	1.66	1.76	1.63	2.48	2.56	1.91
GSM8K peak	318	275	275	318	275	275
Game24 acc.	0.375	0.500	0.562	0.562	0.562	0.625
Game24 e2e (s)	1.61	1.64	1.13	2.15	2.15	1.52
Game24 peak	265	216	216	265	216	216

Table 11. Expand versus decode, milliseconds per problem. Expand is the prefill of the k residuals across three turns. On APC that timer also includes the two 32-token step decodes; ForkServe and ForkServe+ charge those steps to decode. Speedup in Figure 9 is APC expand time over the other system. Shortest time in each trio in green.

	Expand (ms)			Decode (ms)		
	APC	FS	FS+	APC	FS	FS+
4B GSM8K	429	83	55	1215	1680	1572
4B Game24	435	87	53	1166	1552	1074
8B GSM8K	597	94	58	1867	2468	1848
8B Game24	598	94	56	1538	2060	1466

4B (15/16 on the same problems) and is one problem lower on Qwen3-8B (15/16 vs. 16/16). On Game24, ForkServe solves two more problems than APC on Qwen3-4B (8/16 vs. 6/16) and the same nine on Qwen3-8B; ForkServe+ solves 9/16 and 10/16.

4.5 Cross-model forest

Setup. Same four systems, $n=40$, $D=512$, $tp=2$ and 4, on Qwen3-4B, Qwen2.5-Math-1.5B, Qwen3-14B, Llama-3-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Llama-8B.

Peak occupancy. APC retains the shared trunk and the union of the residuals, which is the copy-on-write term $M_{APC} = M_{CoW} = b(L + \sum_i \ell_i)$ from (1). The tree retains only the committed spine $M_{spine} = b(L + \ell_*)$. The ratio of these two terms cancels b and equals $(L + \ell_*) / (L + \sum_i \ell_i)$, so it depends on k and the lengths (L, ℓ_*) rather than on the weights or the tensor width. When the residuals have similar length, the ratio simplifies to $(L + \ell) / (L + k\ell)$ and decreases as L/ℓ decreases. Figure 7 shows this pattern: a long trunk on GSM8K gives 0.72–0.75, and a shorter trunk on Game24 gives 0.57–0.61. Figure 10 repeats the comparison at $n=80$ with recompute and ForkServe+ on separate axes. Relative to recompute, which stores the cloned occupancy M_{clone} , the spine is 0.25 $\times$. ForkServe+ matches that ratio. Accuracy remains at the APC level, which indicates that the lower peak is a storage effect on the same committed answer. Figure 11 shows the per-item decode-length distribution on GSM8K and Game24.

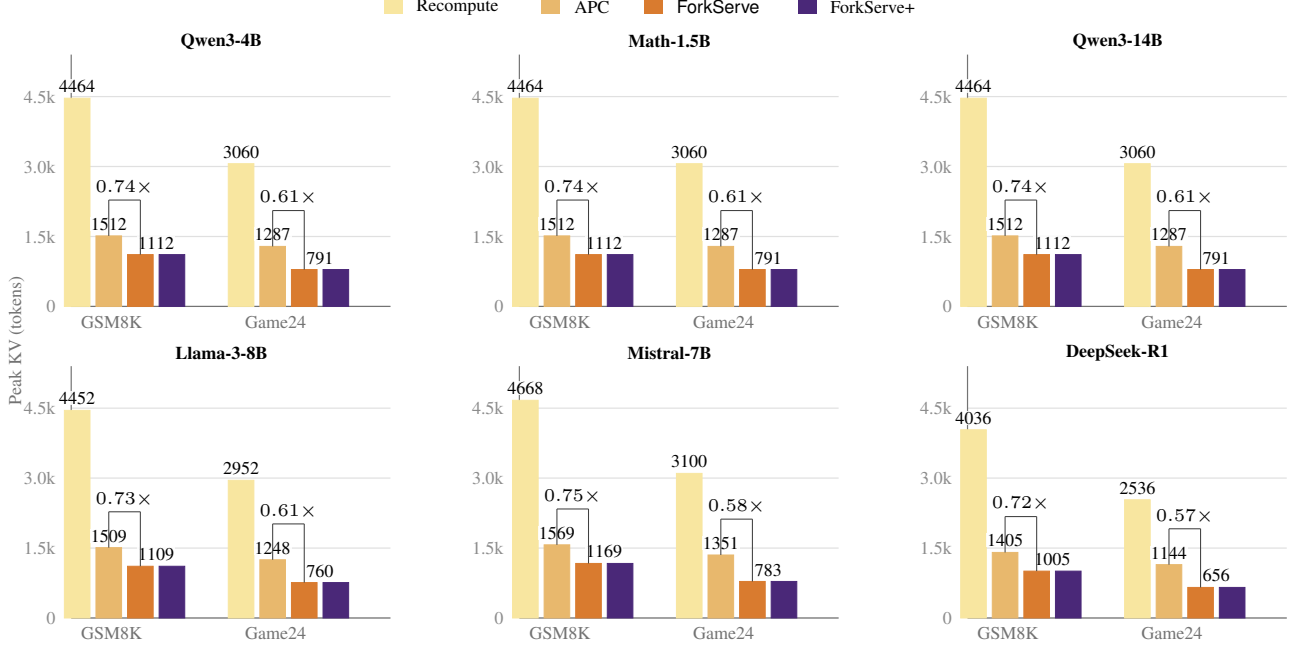


Figure 7. **Peak KV** on six models ($n=40$, $tp=2$). Each panel is one family; groups are GSM8K and Game24. Bars are recompute, APC, ForkServe, and ForkServe+ (cream / tan / orange / purple). ForkServe and ForkServe+ share the spine. Annotated $\times$ is $M_{\text{spine}}/M_{\text{APC}}$: 0.72–0.75 $\times$ on GSM8K and 0.57–0.61 $\times$ on Game24, independent of checkpoint size, as (1) predicts.

Table 12. Last-line answer stop, $n=32$, $tp=2$. ForkServe+ stops only on a finished answer in the last non-empty line. Peak is the max over chunks. GSM8K $D=512$; Game24 $D=256$. R1-8B is DeepSeek-R1-Distill-Llama-8B; L8B is Llama-3-8B; M7B is Mistral-7B. Best storage in green.

Model	Metric	APC	FS+
R1-8B	GSM8K peak	1493	1093
	GSM8K acc.	0.500	0.562
	GSM8K D^+	16384	12693
	Game24 peak	1064	616
L8B	GSM8K peak	1485	1085
	GSM8K acc.	0.719	0.688
	GSM8K D^+	16384	5101
	Game24 peak	1224	736
M7B	GSM8K peak	1521	1121
	GSM8K acc.	0.344	0.188
	GSM8K D^+	16384	6723
	Game24 peak	1296	728

HumanEval uses $k=2$, so the residual is a smaller fraction of the context and M_{spine} remains close to M_{APC} . Pass@1 is unchanged across recompute, APC, and ForkServe.

4.6 Branching, pruning, and long traces

Setup. GSM8K, $n=16$, $D=512$, $tp=2$, on Llama-3-8B, Mistral-7B, and DeepSeek-R1-8B. The sweep changes k or adds a second ToT level. MATH-500 ($n=200$) and AIME ($n=73$) use $D=2048$.

Table 13. GSM8K peak KV versus branching ($n=16$, $tp=2$, $D=512$). The spine peak is invariant in k ; the APC peak increases with k . ratio is FS/APC. L8B is Llama-3-8B; M7B is Mistral-7B; R1-8B is DeepSeek-R1-Distill-Llama-8B. Best storage in green.

Model/ratio	Method	$k=2$	$k=4$	$k=8$	depth 2
L8B	APC	610	750	998	814
	FS	550	550	550	614
	FS/APC	0.90	0.73	0.55	0.75
M7B	APC	629	769	1033	841
	FS	569	569	569	641
	FS/APC	0.90	0.74	0.55	0.76
R1-8B	APC	629	761	1053	841
	FS	561	561	561	641
	FS/APC	0.89	0.74	0.53	0.76

The ratio falls with k . Table 13 and Figure 12 measures $(L+\ell)/(L+k\ell)$ on the GPU. The spine peak is invariant in k while the APC peak increases, so the ratio moves from 0.89–0.90 at $k=2$ to 0.53–0.55 at $k=8$ on all three families. At depth 2 the ratio is 0.75–0.76. ForkServe+ leaves the peak unchanged. At $k=8$ it rejects 28 branches and reduces fan-out from 307 to 179 ms on Llama-3-8B, from 349 to 191 ms on Mistral-7B, and from 363 to 194 ms on DeepSeek-R1. With draft pruning disabled, Llama-3-8B fan-out is 247 ms and DeepSeek-R1 fan-out is 271 ms. A batch of 16 keeps the ratio at 0.72–0.73 (2071/2871 on Llama-3-8B, 2135/2935 on Mistral-7B, 2093/2893 on DeepSeek-R1).

At $D=2048$ the ratio is 0.75–0.91 (Table 14). On MATH-

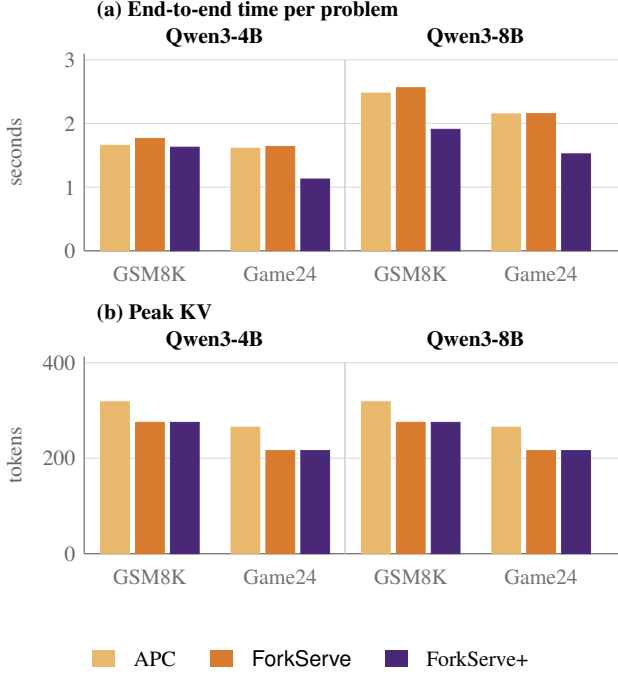


Figure 8. **Three-turn ToT** on Qwen3-4B and Qwen3-8B ($n=16$, $tp=2$, one problem per generate). (a) End-to-end seconds per problem; a shorter bar is faster. (b) Peak KV tokens. ForkServe stays with APC on latency and stores the spine ($0.86\times$ APC on GSM8K, $0.82\times$ on Game24). ForkServe+ is faster because it prefills one of four residuals.

500, ForkServe matches APC (432 s vs. 429 s on DeepSeek-R1). ForkServe+ cuts DeepSeek-R1 from 429 s to 247 s and Qwen3-4B from 329 s to 236 s; the Qwen3-4B $n=32$ slice is 38 s versus 53 s. ForkServe+ cuts MATH-500 fan-out from 2406 to 1284 ms on DeepSeek-R1 and from 2787 to 1014 ms on Qwen3-4B, against APC at 1593 and 1201 ms. Qwen3-4B ForkServe+ scores 120/200 on MATH-500 (APC 124/200).

5 RELATED WORK

Request-level serving. Orca [47] introduced iteration-level scheduling; vLLM [18] paged the KV cache and hashes full blocks for APC; SGLang [53] added radix prefix sharing; Sarathi [2, 3] chunked prefills; DistServe and Splitwise [54, 31] disaggregated prefill from decode. Prefix caching discovers sharing after tokens exist. Prefill-decode disaggregation isolates TTFT from inter-token latency; it does not select which branch to compute, and it transfers every page the prefiller produced. APP uses both mechanisms as filters: the hash as a skip, the connector as a gate.

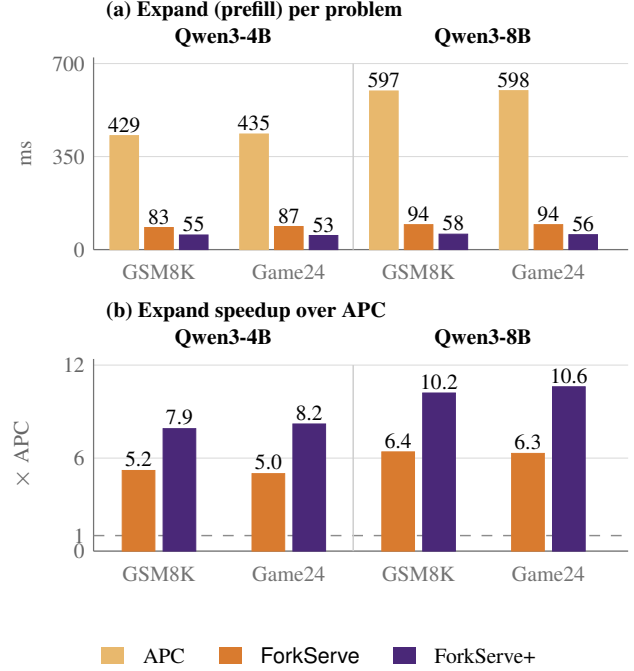


Figure 9. **Expand versus APC** on the same four runs as Table 11. (a) Expand milliseconds per problem. (b) Speedup of that expand over APC. The dashed line is $1\times$. ForkServe is $5.0\text{--}6.4\times$ and ForkServe+ is $7.9\text{--}10.6\times$. Decode in Table 11 does not repeat that ratio: the step decodes sit in the ForkServe decode column, so ForkServe end-to-end stays with APC.

Program- and SLO-aware scheduling. Autellix [25] schedules by program-level attained service; JITServe [52] allocates just-enough bandwidth under heterogeneous SLOs. Hermes [24], ThunderAgent [17], and SAGA [14] raise the atom to the program or workflow. These systems leave KV unforked and unissued children unpreilled.

KV retention and overlap. InferCept [1], Continuum [20], MORI [43], and ConServe [11] keep the *current* context alive or placed. Sutradhara [6] overlaps tool execution with prefill of one linear tool-independent prefix. SmoothAgent [30] lookahead-transforms context; IdleSpec [9] drafts plans for accuracy, not KV. Classical speculative decoding [19, 7, 22] drafts *output* tokens. ForkKV [38] CoW-splits LoRA residuals. We represent the future as a *set* of CoW children, speculate several known suffixes, commit by LCP, prune losers before prefill, and gate P/D insert.

Decoding-time pruning. A second line drops a candidate after the model has generated it. Beam search in Tree-of-Thoughts [45] scores a finished step and keeps the top beams. Early-stopping self-consistency (ESC) [21] draws a window of complete samples and skips later samples only once that

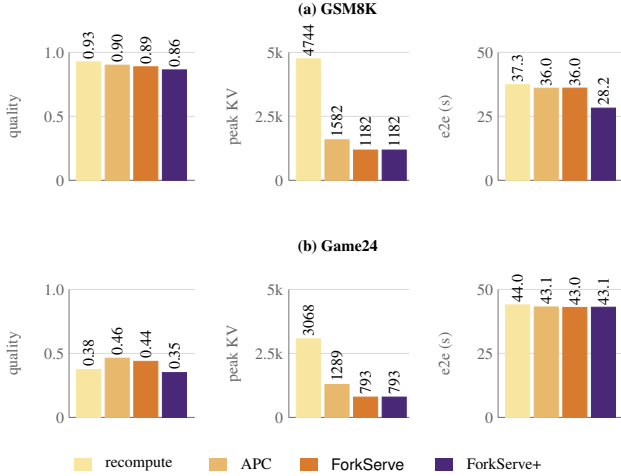


Figure 10. Four systems on Qwen3-8B ($n=80$, $tp=2$, $D=512$). Each panel has its own axis. Quality is higher-better; peak KV and end-to-end time are the measured values, so a shorter bar is better. (a) GSM8K. (b) Game24. Figure 10 shows that ForkServe and ForkServe+ lower peak KV below APC on both workloads ($0.75\times$ on GSM8K, $0.62\times$ on Game24) while GSM8K quality stays with APC, and that ForkServe+ also lowers GSM8K end-to-end time from 36.0 s to 28.2 s.

window agrees. Speculative Rejection [35] ranks partial generations and halts the lower quantile. Dynamic Parallel Tree Search (DPTS) [12] generates a fixed mini-step on every active node, then early-stops low-confidence children. In each method the score is computed from tokens already resident in the KV cache, so the prefill of the candidate and a decode prefix are issued before the drop. APP scores a known residual first and withholds that prefill, which also keeps the child off the disaggregated connector (§4.2). The same τ_{pre} sits in front of ESC, Speculative Rejection, or DPTS (§4.2).

6 CONCLUSION

The KV retained for aborted branches limits the throughput and memory cost of agentic LLM serving. A Tree-of-Thoughts fan-out shares one trunk, yet a prefix cache retains the residuals of that expand, and a disaggregated connector transfers those pages when the harness later aborts them. We introduced ForkServe, which makes the branch a serving object: a copy-on-write context tree, a cascade that rejects a residual at the minimum-cost stage at which rejection is admissible, and a decode path that attends only the committed spine.

Our evaluation demonstrates a $0.72\times$ median peak KV on Qwen3-14B GSM8K at matched accuracy (0.845 vs. 0.844), a 74.72% cut in modeled fan-out prefill, and a

Table 14. Contest forest, $D=2048$, $k=4$, $tp=2$. MATH-500 is $n=200$; AIME is $n=73$. R1 is DeepSeek-R1; 4B is Qwen3-4B. Best storage and shortest end-to-end time in green.

Model	Method	peak	/ APC	acc.	e2e (s)
R1 MATH	APC	1863	—	0.560	429
	FS	1391	0.75	0.510	432
	FS+	1599	0.86	0.545	247
R1 AIME	APC	2118	—	0.055	178
	FS	1646	0.78	0.123	178
	FS+	1662	0.78	0.096	165
4B MATH	APC	1937	—	0.620	329
	FS	1770	0.91	0.625	248
	FS+	1770	0.91	0.600	236
4B AIME	APC	2150	—	0.027	136
	FS	1794	0.83	0.041	136
	FS+	1794	0.83	0.041	131

35.53% increase in the concurrency a fixed pool can sustain. Across six families the GSM8K spine stays at $0.72\text{--}0.75\times$ that peak, and the ratio falls from $0.89\text{--}0.90\times$ at $k=2$ to $0.53\text{--}0.55\times$ at $k=8$. At a 2048-token budget the ratio remains $0.75\text{--}0.91\times$, and ForkServe+ MATH-500 fan-out is 1284 ms against the prefix cache at 1593 ms on DeepSeek-R1 and 1014 ms against 1201 ms on Qwen3-4B. On Qwen3-8B, the same contest slice at $n=32$ and $D=1024$ shows why a blanket draft skip is the wrong rejection: dropping every residual below τ cuts any-finished accuracy from 20/32 to 13/32 at $k=16$, while a 128-token probe that continues only if the generated prefix scores at least τ restores 20/32 and lowers end-to-end time from 104.1 s to 72.2 s. On GSM8K ($n=16$, $D=512$), $\tau_{pre}=0.15$ is the accuracy-preserving point: 15/16, 16/16, 16/16 at $k=4, 8, 16$ and -5% , -6% , -17% end-to-end against native ESC, with a 23% cut in computed prefill on ESC, Speculative Rejection, and DPTS alike. $\tau_{pre}=0.45$ is the extra-speed point (-36% at $k=16$, still 16/16) and loses two questions at small fan-out (14/16).

ForkServe’s key innovations include: (i) a copy-on-write context tree that turns `fork` into an alias of the parent’s KV pages and frees only the residual on abort, so live occupancy is the spine $b(L + \ell_*)$; (ii) advanced prefill pruning, which draft-skips repeated loops at $\tau_{pre}=0.15$ before any host decoder runs, then a probe that extends any other low-score residual only when the generated prefix clears the decode threshold, and a transfer gate that inserts only the miss tail of a residual that is kept; (iii) longest-common-prefix commit, which keeps speculative tokens out of the sampler so the committed sequence matches a prefill of committed tokens only; and (iv) a two-class scheduler that admits speculative prefill only from the slack left by committed requests.

Future directions include probe lengths that track fan-out width, closed-loop QPS sweeps of the analytical concurrency gain, full-set Game24 and HumanEval forests, deeper trees and 32K-context windows, and multi-tenant admission with explicit latency margins on a shared frame pool. ForkServe shows that branch cost is incurred only for the

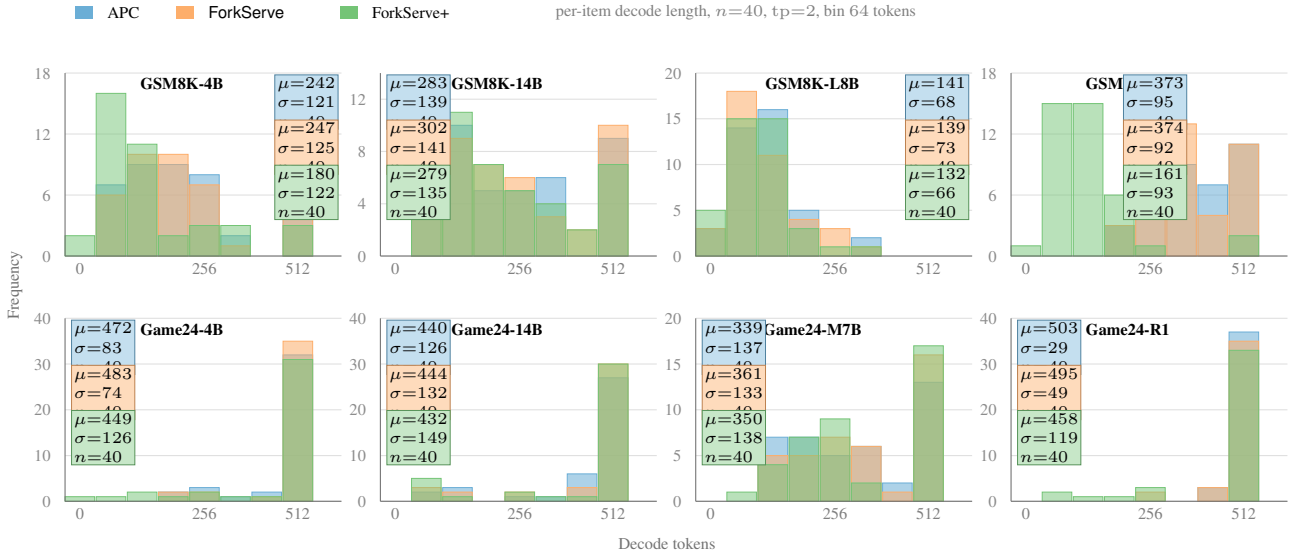


Figure 11. Per-item decode-length histograms ($n=40$, $tp=2$, $D=512$). Each panel overlays APC, ForkServe, and ForkServe+ (bin width 64). Boxes report μ , σ , and n . GSM8K (top) and Game24 (bottom) overlay APC, ForkServe, and ForkServe+. Llama-3-8B Game24 is omitted (all lengths 1); Mistral-7B is shown instead.

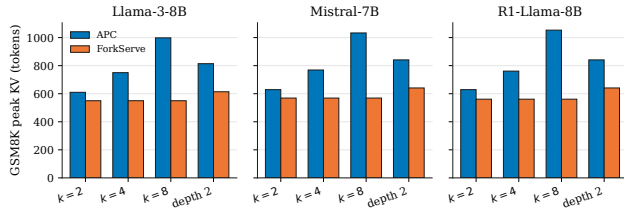


Figure 12. GSM8K peak KV versus branching ($n=16$, $tp=2$, $D=512$). ForkServe+ matches the ForkServe bar. The spine peak is invariant in k ; the APC peak grows with the hashed union, so the ratio falls from 0.89–0.90 to 0.53–0.55.

residual that is committed, and only at the minimum-cost stage in the rejection cascade: the context tree and prefill admission together yield a spine whose occupancy is set by k and residual length, independent of tensor-parallel width. As agentic workloads scale, fan-out width, tree depth, and context length all grow against a fixed KV pool. Serving systems that fork by reference and prune before prefill retain capacity on the committed path because each new branch aliases the trunk, a residual is rejected at the first admissible stage, and abort returns only that residual, so the tokens that occupy HBM are the spine the decoder attends.

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